

Incentives, Management, and Public Service Delivery: Evidence from a large field experiment in Peru*

Sarita Oré-Quispe

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Abstract

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1 Introduction

Globally, subnational governments account for 21.5% of public expenditure and 8.3% of GDP, delivering essential public services such as education, health, water, and sanitation (OECD, 2022). Central government transfers fund over half of subnational revenues (OECD, 2022), making transfer design a primary lever to shape the quantity and quality of local service provision. That recognition has pushed governments increasingly toward performance-based transfer schemes,¹ raising a fundamental question in public sector contract design: should contracts reward verifiable service delivery inputs that local agents perform, or directly reward the final outputs the principal ultimately cares about?

Input-based contracts reward tasks directly under the agent's control, shielding them from external risk, but they translate into output gains only when those specific inputs constitute the binding constraint and risk crowding out non-incentivized quality margins (Mohanalan et al., 2021; Grant Miller et al., 2014; Holmström et al., 1991). Conversely, output-based contracts align incentives with the principal's objective and grant agents the flexibility to deploy local information, but outputs are often noisier signals of agent effort (Baker, 1992; Baker, 2002). This paper asks whether adding an output indicator to an input-based transfer contract can strengthen local management, service delivery, and ultimately the outputs the principal cares about, and whether these changes persist once the output indicator is removed. I answer these questions with a large field experiment across 539 rural municipalities in Peru, representing 85% of all rural municipalities in the country.

In rural Peru, many infants miss the preventive care that protects them in their first year of life. Routine checkups allow health staff to catch developmental delays and illness before they become serious, and vaccines prevent the leading infectious causes of child death [CITE]. Yet in 2023, only X% of rural infants under 12 months completed their scheduled checkups, and X% lacked their scheduled vaccines (ENDES, 2023). Since health facilities also distribute iron supplements, infants who miss visits also miss the main protection against anemia, the most visible sign of poor child nutrition. Only X% of rural infants aged 6 to 11 months took iron supplements and 64.9% were anemic (ENDES, 2023).² In 2019, the central government created the Municipal Home

¹Over the past two decades, at least 84 countries have adopted performance-based transfers in education and health (Muralidharan et al., 2020).

²Iron deficiency is the leading cause of childhood anemia, so policy has long focused on distributing iron supplements through health facilities [CITE]. Supplementation improves hemoglobin only if families pick up the supplement and give it to the child at the correct dose and frequency, and if the child's intestine absorbs the iron, a process that diarrhea and intestinal parasites can impair [CITE]. Under routine conditions, only 24.4% of children

Visit Program and tied its implementation to the *Incentive Program for Municipal Management Improvement* (hereafter, Incentive Program), which conditions transfers to municipalities on their performance. Since then, the Incentive Program has rewarded municipalities for home visits that promote preventive health care and supplementation. Although coverage reached about 90%, service use stayed low and anemia high, suggesting that home visits reached families leading them to seek care at the health facility.

To test whether rewarding outputs shifts managerial effort from the coverage of home visits toward their effectiveness, I partnered with the Ministry of Economy and Finance and the Ministry of Health. We randomly assigned 270 municipalities to the status quo input-based contract and 269 to a mixed contract. The input contract rewards the share of young children and pregnant women who receive home visits.³ The mixed contract adds an output indicator, the share of children who are not anemic or whose hemoglobin does not decrease.⁴ No arm operates without incentives. The experiment varies what the contract rewards while holding fixed how much it pays. The contracts ran from August to December 2024. From January to May 2025, an unanticipated administrative decision reverted all municipalities to the input-only contract. This reversion provides an special feature to test whether changes persist once the output indicator is removed. I measure managerial practices with manager surveys before and after the trial. Additionally, to ensure that these results are not driven by reporting bias or experimenter demand effects, I complement this information with an endline survey of health facility, who are not part of the incentive scheme and provide third-party reporting. I measure service utilization and child health with administrative records that cover every child's visit to a public health facility.

Each municipality appoints one manager to run the Home Visit Program. On average, each manager oversees 11 mission-motivated community health workers (CHWs), who receive only a transport stipend to reach households (Besley et al., 2005). Each CHW visits up to 15 families, up to three times a month. During these visits, CHWs deliver standardized messages on preventive care, nutrition, and iron supplementation, encourage families to attend their scheduled checkups, and refer children at risk of anemia to the local health facility. CHWs cannot provide medical

reach 90% adherence [CITE].

³Because anemia originates in maternal iron deficiency during pregnancy and low iron intake in the first months of life, prevention targets pregnant women and infants.

⁴A mixed contract of this kind is a natural middle ground, since it preserves the insurance that input indicators give local governments while tying part of the reward to the principal's objective. The output indicator counts a child as a success if hemoglobin does not fall between the 6- and 12-month screening tests, even if it does not rise. This definition reflects infant physiology. The iron stores a child accumulates before birth run out by about six months, just as rapid growth raises iron needs, so without adequate iron intake hemoglobin tends to decline over this period (Cerami, 2017).

care or dispense supplements, so families must obtain iron at the health facility. Health facility staff also train CHWs and treat the children CHWs refer. The manager links the incentive contract to this frontline work.⁵ Because CHWs have no employment relationship with the municipality, managers cannot pass monetary incentives down to them. Any response to the output indicator must therefore work through non-monetary practices, such as closer monitoring, better coordination with health facilities, and recognition of CHWs' work. At baseline, 87% of managers report high or medium autonomy over program implementation. How much of the transfer becomes effective service delivery therefore depends largely on how managers use this discretion.

The government announced each municipality's assignment through official resolutions and workshops, and I complemented these with a nationwide campaign that sent each municipality its own indicators and respective thresholds. By October 2024, 98.2% of managers correctly identified their contract type. I present four main results. First, the mixed contract did not expand home visit coverage. Under the input-only contract, visits already reached 88% of targeted children, and the mixed contract changed coverage by 0.5 percentage points (p.p.). Coverage of pregnant women, near 94% in both arms, did not change either.

Second, the mixed contract strengthened managerial practices without adding resources. Managers reorganized their work around the output indicator. Managers under the mixed contract were 13.7 p.p. (19%) more likely to track the anemia indicator daily or biweekly, but no more likely to verify CHWs' visits directly. They also tightened referral management. They were 12.4 p.p. (24%) more likely to strengthen care for children at risk of anemia. Health facility staff report that managers under the mixed contract were 19.5 p.p. (83%) more likely to coordinate referrals with them directly. According to the same staff, CHWs were 14.4 p.p. (55%) more likely to accompany mothers to the health facility. Managers were also 17.1 p.p. (46%) more likely to reward CHWs with non-monetary rewards. Municipalities under the mixed contract deployed no more CHWs and spent no more on health. The contract reallocated managerial attention rather than expanding inputs.

Third, these practices increased the use of preventive health services but did not improve child health. Children in mixed-contract municipalities made 0.37 more health facility visits (3.9%) and received 0.28 more vaccine doses (8.8%). Iron supplement pick-ups nearly doubled,

⁵The mayor receives the transfer on behalf of the municipality and oversees the manager. At baseline, 80% of managers reported that missing the contract targets would affect their professional standing, which gives them a personal stake in the contract.

rising by 0.36 from an input-only mean of 0.36. Hemoglobin and anemia prevalence, however, did not change. The hemoglobin estimate is -0.03 g/dL (s.e. 0.07), which rules out gains larger than 0.10 g/dL. This bound is less than one-fifth of the 0.56 g/dL gain that clinical trials with enforced iron adherence achieve (Gera et al., 2007). The estimates also rule out reductions in anemia prevalence larger than 3.6 p.p. The null does not reflect insufficient exposure. The main sample includes children with at least one month of exposure to the contract. Estimates remain unchanged for children with at least three months of exposure, the horizon at which clinical trials find most of the hemoglobin gain. Following the same cohort over a longer window also yields no effects, so the null is not a matter of timing. Nor does the null seem to reflect an environment that blocks absorption.⁶ I interpret this null as evidence about how effort maps into hemoglobin, not as a failure of the incentive to reach families. Picking up iron is only the first link in the chain. Hemoglobin improves only if the child takes the supplement daily and the children's body absorbs it. Similarly, I detect no effects on nutritional status as measured by weight-for-age z-scores (WAZ).

Fourth, the managerial changes outlasted the contract. From January to May 2025, all municipalities reverted to the input-only contract. Managers in formerly mixed-contract municipalities remained 13.4 p.p. more likely to track the anemia indicator, nearly the same effect as during the trial. They kept coordinating referrals with facility staff (10.3 p.p.) and rewarding CHWs (13.4 p.p.), and their CHWs kept accompanying mothers (11.3 p.p.). Only the small gain in training faded. These managers also served a new cohort of children, born too late to be reached during the trial. Children in this cohort used health services more, with X additional facility visits and X additional iron pick-ups, but their health outcomes remained unchanged. Because their families never experienced the mixed contract, this persistence cannot reflect habits that households formed during the trial. It points instead to the supply side.

This paper contributes to three main strands of literature. First, it advances the personnel economics of the state and contract design by directly comparing an input-only incentive scheme against a hybrid input-plus-outcome contract within an institutional transfer framework. While a rich literature evaluates individual performance incentives for frontline public workers (Muralidharan et al., 2011; Duflo et al., 2012; Ashraf et al., 2014) as well as institutional transfers to subnational units, schools, and health facilities (Lavy, 2002; Basinga et al., 2011; Gertler et al.,

⁶Poor water and sanitation expose children to intestinal infections and parasites, which cause inflammation and reduce iron absorption. If absorption were the binding constraint, effects should appear where water and sanitation are better. Effects do not differ by access to water and sanitation, or by baseline poverty.

2012; Olken et al., 2014; Celhay et al., 2019; Luo et al., 2020; Bandiera et al., 2021), empirical evidence directly contrasting input- versus outcome-based incentives in the same institutional setting remains remarkably scarce (Fryer, 2011; Mohanan et al., 2021). I fill this gap by holding the total potential bonus pool constant and isolating how adding an outcome indicator alters administrative choices and service provision.

Second, the paper contributes to the public sector management literature by opening the "black box" of bureaucratic behavior and evaluating the persistence of management practices once temporary incentives expire. A growing body of work establishes that management quality is a major determinant of state capacity and public sector productivity (Bloom et al., 2007; Rasul et al., 2018; Fenizia, 2022). However, civil service rigidities often restrict standard personnel tools such as hiring, firing, or individual merit pay (Bandiera et al., 2021; Rasul et al., 2021; Otero et al., 2025). I show how institutional performance transfers can activate alternative micro-managerial levers—such as active target monitoring, inter-institutional coordination, and worker recognition—and provide rare evidence on whether these activated practices persist after the fiscal contract reverts to the status quo (Celhay et al., 2019; Dunsch et al., 2023).

Finally, this study adds to the literature on health service delivery and Community Health Worker (CHW) interventions in low- and middle-income countries. Prior research on CHWs primarily focuses on direct worker-level incentives through financial compensation (Deserranno, 2019; Björkman Nyqvist et al., 2019), social recognition (Ashraf et al., 2014; Bandiera et al., 2022; Fracchia et al., 2023), or career progression (Ashraf et al., 2016). In contrast, I demonstrate that incentivizing local government institutions serves as a powerful top-down policy lever to effectively mobilize frontline volunteer health workers and expand primary care delivery without altering individual worker-level compensation.

The rest of the paper proceeds as follows: Section 2 provides the institutional context, describing Peru's municipal incentive structure and the prevalence of childhood anemia. Section 3 details the Home Visit Program. Section 4 describes the experimental design. Section 5 presents the main results. Finally, Section 6 concludes.

2 Background

This section describes the institutional and public health setting of the study. It first describes Peru's administrative structure and the Municipal Incentive Program, a national performance-

based scheme linking performance transfers to service delivery targets. It then documents the prevalence and consequences of childhood anemia in Peru.

2.1 Administrative organization and municipal incentives

Administratively, Peru is divided into 25 regions, 196 provinces, and 1,891 districts. Regional governments oversee the regions, while local governments, or municipalities, administer districts. Municipalities' core functions are basic local services — water and sanitation, waste collection, maintenance of public spaces, local infrastructure, business licensing, and local tax collection. Health is formally a shared competence, but the transfer of functions to municipalities was never completed. Regional governments deliver primary healthcare and hospital services, with little municipal participation (Alves et al., 2025). Municipalities do not run health facilities, employ medical personnel, or deliver medical care; their involvement in health runs through the health sector. They can organize and finance community-based activities, but diagnosis and treatment remain with the health facility.

In 2009, the Ministry of Economy and Finance launched the Incentive Program for Municipal Management Improvement⁷ to improve the quality of local service delivery. The incentive program operates as an institutional performance-based financing scheme, allowing municipalities to access additional budget conditional on achieving specific service delivery targets. The transfer goes to the municipality as an institution, not to the individuals who implement the program. Governments often favor institutional over individual incentives because they limit discretion, enhance transparency, and reduce corruption risks (...). The design has a cost, however: it can generate free-riding and weak internal accountability, since the rewards need not translate into direct benefits for frontline workers (G. Miller et al., 2014).⁸

Participation in the Municipal Incentive Program is voluntary; however, in practice, all municipalities participate because the potential transfer represents an important share of their expected annual tax revenue. On average, municipalities can obtain resources equivalent to 15.7 times their expected annual tax revenue (Figure ??), with a median ratio of 3.4. The incentive payments are disbursed after performance verification, typically during the following fiscal year, and can be used freely to finance the provision of public services.

⁷In Spanish, *Programa de Incentivos a la Mejora de la Gestión Municipal*. Before 2016, it was known as *Plan de Incentivos a la Mejora de la Gestión Municipal*. For more information, see www.mef.gob.pe.

⁸Performance-based transfers of this kind have become increasingly common across Latin America and the Caribbean (Alves et al., 2025). A survey of 17 Latin American and Caribbean countries finds that performance-based financing is in place in 14 of them (Pimenta et al., 2023).

2.2 Children's anemia in Peru

Anemia is a condition in which the number of red blood cells or the hemoglobin concentration is lower than normal. Anemia during early childhood is broadly associated with growth retardation, reduced resistance to infections, delayed psychomotor development, lower language acquisition, and impaired cognitive and socioemotional development (INS, 2024). These developmental deficits can become a significant deterrent to human capital accumulation in developing countries, and hence a potentially important barrier to economic development (Chong et al., 2016).⁹

Yet anemia continues to be a pressing global concern. Nearly half of all children under five worldwide are affected (42.8%), and the burden is even more severe in low- and middle-income countries, where prevalence reaches 56.5% (WHO, 2025). In Peru, the prevalence of anemia among children under five years of age is 33.4%, and 60.5% among those aged 6 to 12 months (INEI, 2023). Prevalence is consistently higher in rural areas where households are poorer and have less access to basic services: among children aged 6 to 12 months, it reached 64.9% in rural areas in 2023, compared to 58.7% in urban areas. Rural households are also poorer and have less access to basic services.¹⁰

Anemia is primarily caused by iron deficiency (WHO, 2023) and it is the most prevalent micronutrient deficiency in the world (Chong et al., 2016). For children under one year of age, the most appropriate interventions are iron supplementation and caregiver nutritional education, given their age-specific nutritional needs and feeding practices.¹¹ Supplementation is administered daily in drops or syrup for at least three consecutive months, and adherence is critical to its effectiveness (WHO, 2016). Clinical trials show that when adherence is enforced, supplementation can raise hemoglobin by 0.56 g/dL among children under 24 months (Gera et al., 2007).

Although iron deficiency is the leading cause, anemia is a multi-causal condition. Beyond dietary iron intake, hemoglobin status in early childhood depends on maternal iron stores transferred in utero (Cerami, 2017) and children's capacity to absorb ingested iron, which is

⁹Chong et al. (2016) find that, among peruvian teenage students who were anemic at baseline, three months of iron supplementation raised average test scores by 0.4 standard deviations and increased educational aspirations by 16 percent.

¹⁰36% of rural households live in poverty or extreme poverty, compared to 15% in urban areas; 85% have access to sanitation, compared to 96%; and 90% rely on government-sponsored health insurance, compared to 59% (ENAH0, 2024).

¹¹The World Health Organization recommends four strategies to combat anemia: iron supplementation, food fortification, nutrition education, and dietary diversity (WHO, 2006). Although food fortification is generally considered the most effective and cost-efficient strategy (Mithra et al., 2020), it is poorly suited to this age group. Infants are typically exclusively breastfed until six months, and even after complementary feeding begins, small portion sizes limit the intake of fortified and diverse foods.

shaped by environmental conditions such as access to safe water and adequate sanitation (Kothari et al., 2019; Larsen et al., 2017).¹² Other micronutrient deficiencies (folate, vitamin B12, vitamin A) and inherited disorders contribute as well (WHO, 2001). As a result, even sustained iron supplementation need not translate into measurable hemoglobin gains when these complementary conditions are not met.

3 The Home Visit Program

Home visit programs are a widely used service delivery model in low- and middle-income countries. They bring services into the household, reaching families that have little contact with health, education, and social programs. A growing body of evidence shows they improve caregiver knowledge (Stansert Katzen et al., 2020), feeding practices (Galasso et al., 2019; Janmohamed et al., 2020), and child development (Yousafzai et al., 2014; Grantham-McGregor et al., 2020; Zhou et al., 2023). Peru adopted this model against childhood anemia in 2019, when the Municipal Incentive Program began rewarding municipalities for implementing a new home visit program, *Compromiso 1*. The program seeks to increase adherence to iron supplementation among children under 12 months of age.¹³

The Home Visit Program is a community education initiative that delivers information to caregivers on how to prevent and reduce anemia in children. The program targets 879 municipalities, including all 245 urban municipalities located in main cities and 634 of the most rural municipalities (Figure ??).¹⁴ Municipalities, in collaboration with healthcare staff, conduct home visits to eligible households. The program targets reaches the most vulnerable populations: in urban districts, the program targets low-income families; in rural districts, coverage is universal.¹⁵

¹²Diarrhea and intestinal parasites cause infection and inflammation that impair iron absorption, so a child can take the full dose and still absorb little of the iron (Pasricha et al., 2021).

¹³The Ministry of Health designed the program. Its full name in Spanish is *Compromiso 1: Mejora del estado nutricional y de salud de las niñas y niños de 3 a 12 meses*. It builds on *Cuna Más*, a home visit program run centrally by the Ministry of Development and Social Inclusion since 2012. *Compromiso 1* instead assigns delivery to local governments. It thus marked a first step toward the decentralization of public health functions, giving municipalities an operational role in preventive, community-based care.

¹⁴Municipalities are classified into seven types (A to G) based on their urban status, whether they are the main district of a province, and their share of urban population (Table ??). According to this classification, the Home Visit Program targets the urban municipalities from main cities and the group of the most rural municipalities.

¹⁵Eligibility in urban districts is determined by health insurance status. Targeted children are those without any health insurance or enrolled in the *Seguro Integral de Salud* (SIS), a government-sponsored insurance program for low-income populations, equivalent to Medicaid in the United States.

Home visits are tailored to the child’s age and health status. Visits for children aged 1–5 months are preventive, targeting a critical developmental window in which neurological development accelerates and iron needs rise sharply (MIDIS, 2018). For children aged 6–12 months, visits may be preventive or recuperative depending on whether anemia has already been diagnosed. Each household receives up to three visits per month, each lasting approximately 20 minutes. During these visits, CHWs deliver standardized messages on nutrition, hygiene, preventive care, and early childhood development.

3.1 Home visit organization

The Home Visit Program operates through a hierarchy of five actors, illustrated in Figure 1. The Central Government designs the incentive contract and disburses the conditional bonus to the Municipality, headed by the Mayor. The Mayor receives the bonus on behalf of the municipality and holds unilateral authority to hire, dismiss, or reassign the Municipal Manager, who runs the Home Visit Program on the ground.¹⁶ The Manager in turn leads the Community Health Workers (CHWs), who deliver home visits to Families with young children. Health outcomes are ultimately realized at the family level, and verifiable outputs—visit coverage and health indicators—feed back to the Central Government to determine the disbursement of the bonus. The rest of this section describes the main actors.

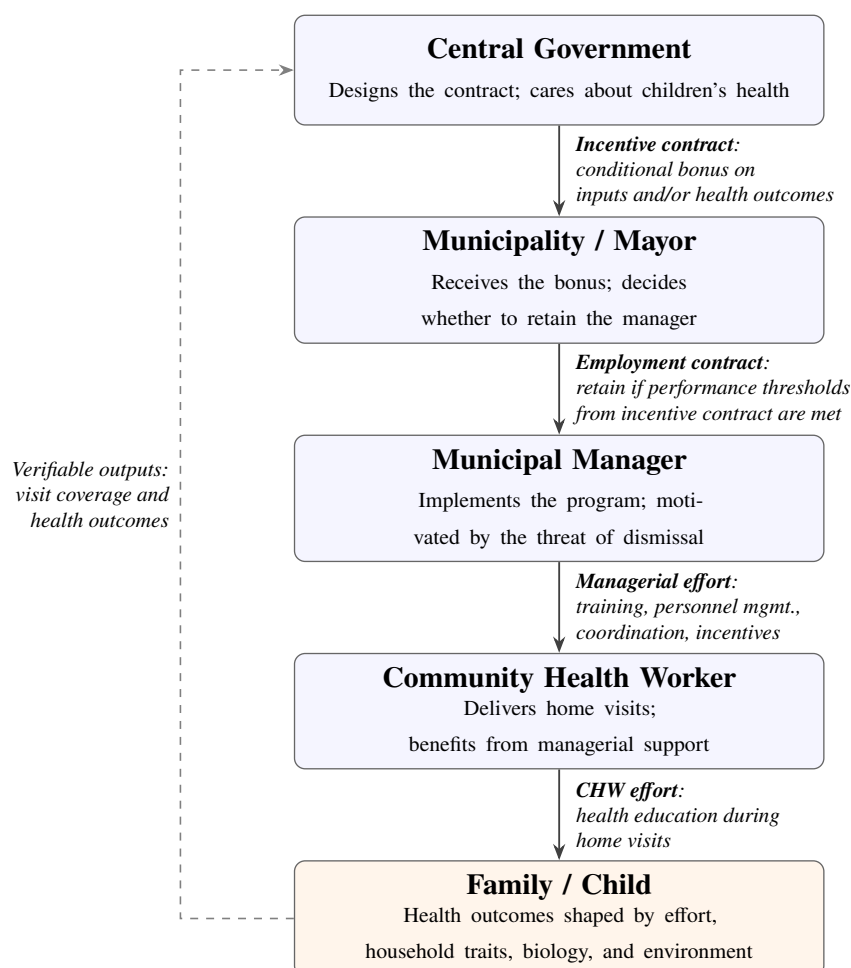
The Municipal Manager

Within each municipality, the Home Visit Program is run by a single Municipal Manager. The manager is appointed by the mayor—who receives the program’s conditional transfer from the central government on behalf of the municipality and retains the authority to dismiss or reassign the manager—and in turn leads the Community Health Workers (CHWs) who carry out the visits.

Managers combine intrinsic and extrinsic motives. Baseline survey data show that 55% are motivated primarily by the program’s contribution to population wellbeing, while the remaining 45% are motivated primarily by meeting the targets that secure the incentive payment. Institutionally, the manager’s key private stake is continued employment: the mayor holds unilateral hiring and dismissal authority, and 80% of managers report that failing to meet the

¹⁶The majority of service delivery agents (80%) are CHWs, so, for simplicity, I use the terms service delivery agents and CHW interchangeably. About half of municipalities are large enough—by target-population size—to add an intermediate organizational layer of supervisors between the manager and the CHWs; each supervisor oversees roughly 20 CHWs.

Figure 1: Institutional structure and flow of incentives



Notes: Solid arrows indicate contractual and hierarchical relations from principal to agent at each layer. The dashed arrow represents the verifiable outputs that the Central Government uses to evaluate municipal performance and disburse conditional transfers.

performance thresholds set by the incentive contract would jeopardize their position.

The municipal manager oversees all health and social programs within the municipality, including the implementation of the Home Visit Program. Within the program, managerial effort consists of five practices. *Monitoring* covers both tracking the performance indicators on which the municipality is evaluated and verifying CHWs' visits directly, through accompaniment, household calls, or re-visits. *Case management* runs from identifying children at risk of anemia to acting on their care, which places the manager as the main point of contact with health facility staff and requires following referred children through to treatment. *Motivating CHWs* involves distributing discretionary rewards and cultivating a sense of mission, complementing the pro-social motives that already drive them. *Training* involves organizing refresher sessions,

coordinated with health facility staff, beyond the mandatory initial training. Managers also hold personnel (i.e., hiring, dismissals, promotions) and budget authority.

Managers exercise substantial discretion over how they allocate effort across these activities: 46% report a high level of overall autonomy, 41% a medium level, and only 11% a low level. As Figure ?? shows, autonomy is highest in coordinating with health facilities (71%) and training CHWs (59%), and remains substantial in hiring CHWs (52%), managing personnel payments (47%), planning activities (44%), and incentivizing CHWs (42%). A large majority (87%) believe the program's success depends on their own effort.

The manager observes the number of home visits each CHW delivers to each child, since visit records are logged in the administrative platform. The manager does not directly observe the quality of individual visits—whether the CHW counseled the mother effectively, assessed the child correctly, and left the family motivated to comply. Quality is inferred indirectly, through occasional accompaniment and through feedback from health facility staff, who verify a random 10% sample of home visits.

The Community Health Worker

CHWs work under a quasi-volunteer arrangement, receiving only a transport stipend of approximately USD 3 per visit that essentially covers commuting costs rather than serving as labor compensation. This arrangement is consistent with the broader CHW workforce in low- and middle-income countries, where engagement relies heavily on pro-social motivations rather than financial returns (Ashraf et al., 2020). To perform their tasks, CHWs receive mandatory two-day in-person training from Health Facility staff on essential topics and learn to recognize children's danger signs for referral. They also receive discretionary on-the-field training from the manager.

CHWs are responsible for scheduling and conducting the home visits.¹⁷ Each home visit involves two distinct margins of effort: quantity and quality. On the quantity margin, the CHW decides how many visits to deliver to each assigned child during the reference period, subject to the program's target of up to three visits per month. On the quality margin, the CHW decides how carefully to follow the visit protocol—that is, how thoroughly to assess the child, how attentively to counsel the caregiver, and whether to refer children showing danger signs.¹⁸ The visit protocol

¹⁷If a caregiver is not home at the scheduled time, the visit is rescheduled rather than skipped; outright refusals are rare, occurring in fewer than 5% of cases.

¹⁸CHWs are not permitted to provide medical care or dispense supplements themselves; when they identify a child at risk, they refer the case to the health facility and encourage the caregiver to bring the child in. Also, by national regulation, CHWs cannot directly dispense iron supplements; medication must be obtained from a health

is standardized through a checklist and a flipchart that guide the delivery of each topic. Family adherence to iron supplementation and healthy practices is influenced by the CHW's effort, but compliance ultimately rests on the family's own choice.¹⁹

The Health Facility

Peru's Ministry of Health classifies health facilities by the complexity of the care they deliver. The categories range from Type I-1 health posts (postas), which offer only basic preventive and outpatient care, to Type III hospitals, which offer specialized treatment and surgery. Type I-1 posts sit at the base of this hierarchy. Type I-1 posts deliver the routine preventive services that children receive during their first year of life—growth-monitoring visits, vaccinations, anemia screening, and iron supplementation through their child health unit. These posts are also the facilities that participate in the Home Visit Program.

Although health facilities are not part of the incentive scheme, they play a key supporting role through three functions beyond direct service provision. First, child health unit staff train CHWs, delivering both the initial mandatory two-day training and discretionary follow-up sessions coordinated with municipalities. Second, they verify the veracity of a random 10% sample of home visits. Third, they manage referrals from CHWs, ensuring that children identified as at risk of anemia or in need of further care are promptly seen and treated.

4 Data

4.1 Administrative data

I draw on four sources of administrative data: the Public Healthcare Information System, which records all health services delivered to children; the Home Visit Program dataset, which tracks CHW characteristics and completed home visits; the Public Budget Information System, which records municipal expenditures and transfers; and district-level poverty and census indicators used to characterize baseline conditions and examine treatment effect heterogeneity.

facility.

¹⁹Compliance is multidimensional—administering iron supplementation daily at the correct dose and time, attending health facility appointments, adopting iron-rich dietary changes, and applying hygiene and sanitation practices—and costly to families in time and effort, particularly in rural areas where health facilities are distant.

4.1.1 Public Healthcare Information System – HIS

I use anonymized administrative data from the public healthcare information system, which records all medical services delivered to children, including diagnoses. These services include routine growth and development check-ups as well as vaccinations. Over the first year of life, a total of 17 vaccine doses are scheduled at specific ages in accordance with national immunization guidelines (Figure ??).²⁰

The database also contains detailed information on anemia screening. Routine hemoglobin tests are conducted at two recommended ages—between 6 and 9 months and between 12 and 13 months—and include measured hemoglobin concentrations as well as the corresponding anemia diagnoses. Children are classified as anemic if their altitude-adjusted hemoglobin concentration is below 10.5 g/dL.²¹ Consistent with this, preventive iron supplementation is provided starting at four months of age and continued during the first year of life. When anemia is detected through routine screening, children are prescribed therapeutic (higher-dose) iron supplementation. Figure ?? summarizes the timing of health-facility–provided services during the first year of life.

The dataset also includes measurements of child weight and height collected during routine visits, from which anthropometric indicators can be constructed. Given the five-month duration of the intervention, I focus on weight-for-age z-scores (WAZ) rather than height-for-age z-scores (HAZ), as weight is more responsive to short-term nutritional changes while length and height reflect longer-term growth processes (Frison et al., 2016).

4.1.2 Home Visit Program

I use the Home Visit Program administrative dataset, which contains detailed information on CHW characteristics, payroll, and the timing of all completed home visits. This database enables the identification of the CHW assigned to each family and allows me to assess CHW turnover. Because CHWs are volunteers receiving only a transport stipend, with no performance pay or career advancement tied to logged visits, the scope for strategic manipulation of these records is limited. This is reinforced by the verification protocol, in which health-facility staff cross-check

²⁰During the first month of life, children attend four growth-monitoring and counseling visits at health facilities, followed by monthly visits through month 11.

²¹Hemoglobin is measured at health facilities using capillary blood obtained via finger prick and analyzed with portable hemoglobinometers. In line with national guidelines and World Health Organization (WHO) standards, hemoglobin concentrations are adjusted for altitude above 500 meters because individuals living at higher elevations typically exhibit higher hemoglobin levels as a physiological response to reduced oxygen availability (WHO, 2011). Without this adjustment, children residing at higher altitudes would be mechanically less likely to be classified as anemic despite facing comparable or higher levels of iron deficiency.

a random 10% sample of visits against household reports.

4.1.3 Public Budget Information System – SIAF

I use administrative data from the *Sistema Integrado de Administración Financiera* (SIAF), Peru's Integrated Financial Management System administered by the Ministry of Economy. SIAF records all public expenditures and transfers executed by national, regional, and local governments, including the funds municipalities receive through the incentive program. The data are available at the monthly transaction level and provide detailed information on expenditure amounts, timing, budget classifications, and revenue sources. I use these data to track municipal spending patterns across functional categories and to assess whether municipalities reallocate resources toward activities related to the Home Visit Program.

4.1.4 District-Level Data

I use district-level poverty indicators from the official **Poverty Map** produced by the *Instituto Nacional de Estadística e Informática* (INEI). The poverty map provides estimates of monetary poverty at the district level using a small-area estimation methodology that combines household survey data with information from the 2017 Demographic Census Data. The incidence of monetary poverty allows me to characterize baseline socioeconomic conditions. In addition, I use district-level indicators of access to safe drinking water and sanitation from the census. Together, these indicators allow me to examine heterogeneity in treatment effects by baseline poverty and environmental conditions.

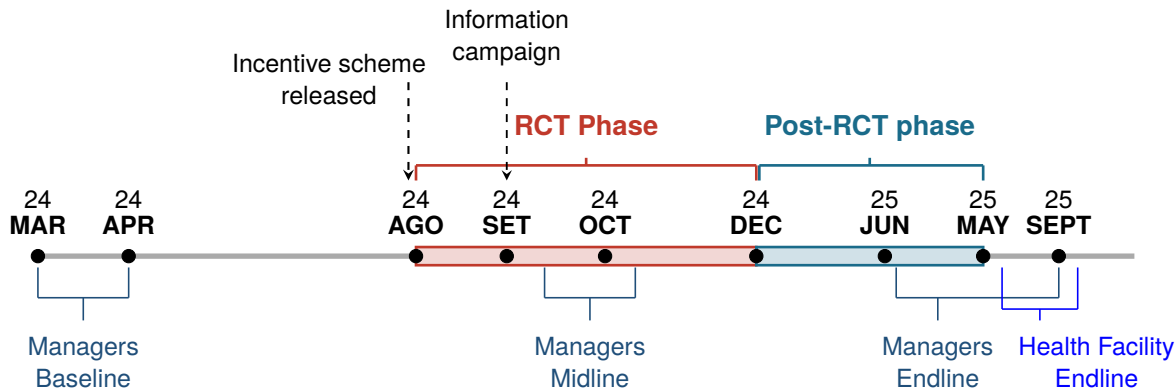
4.2 Survey Data

I collected primary data through two survey instruments, one targeting municipality managers and the other targeting health facility staff.²² The manager survey was administered at three points in time — baseline (March–April 2024), a short midline follow-up (October 2024), and endline (June–July 2025). The health facility survey was administered only at endline (August–September 2025). Endline surveys were designed to retrospectively capture information on two distinct periods: the RCT phase (August–December 2024), when municipalities operated under their assigned incentive contracts, and the post-RCT phase (January–May 2025), when

²²I did not administer a household or caregiver survey, so I do not observe caregiver attitudes, knowledge, satisfaction, or perceived value of the home visits.

an unanticipated administrative delay prevented renewal of the mixed incentive contract and all municipalities reverted to the input-based scheme. Figure 2 summarizes the project timeline. This retrospective two-period structure allows me to assess both the contemporaneous effects and the persistence of any behavioral changes.

Figure 2: Research Timeline



Notes: The figure summarizes the research timeline, including the three rounds of surveys to managers, one round of data collection to health facilities, the release of the incentive scheme in August 2024, the information campaign, and the randomized controlled trial (RCT) implementation period from August to December 2024. Unexpectedly all municipalities reverted to the input-based scheme from January to May 2025. This period is used to assess the persistence of results.

4.2.1 Survey to Managers

Because the municipal incentive operates through municipal managers rather than directly through frontline CHWs, the manager survey is the central instrument for documenting the behavioral mechanisms through which the contract affects implementation. The baseline survey collected data on managers' demographic backgrounds and Home Visit Program staff characteristics. It also elicited beliefs and expectations about the incentive scheme and gathered detailed data on CHW training (e.g., number of sessions and trainers), implemented monitoring practices, incentives to personnel, complementary anemia-reduction strategies, and partnerships with other institutions.

Since the effectiveness of incentive contracts depends on municipalities fully understanding their terms, the midline follow-up assessed managers' comprehension of the incentive scheme and their assigned treatment arm, along with a small set of questions on the implementation of the Home Visit Program. At endline, I successfully tracked 448 managers (83%) who were able to answer questions related to both study periods. Attrition rates were balanced across treatment

arms (see Table ?? and Table ??).²³

4.2.2 Survey to Health Facilities

The endline survey was administered to the health staff member responsible for implementing the Home Visit Program at each health facility.²⁴ The questionnaire collected information on facility-level management practices, supply availability, coordination with municipalities, and verification of home visits. Specifically, it included modules on logistics and availability of equipment, supplies, and iron supplements; coordination mechanisms with municipal teams and CHWs; frequency and modalities of CHW training sessions; and procedures for prioritizing and attending community referrals. Additional sections gathered information on the CHWs' work environment and collaboration dynamics within the health facility.

5 Research design

This section describes the experimental design. I begin by presenting the incentive payment structure and the two contract variants at the center of the experiment. I then document baseline manager motivation and perceived stakes, which shape the behavioral environment in which the contracts operate. Finally, I describe the randomization procedure, the econometric specification, and present checks on incentive awareness and baseline balance.

5.1 Incentive Payment Structure

The incentive program rewards municipalities through a step-function payment schedule defined separately for each performance indicator. For every indicator, the central government sets two pre-determined thresholds: a *partial-bonus threshold* and a *full-bonus threshold*, where the partial-bonus threshold is a lower performance level than the full-bonus threshold. The

²³A potential concern is that the incentive scheme could differentially affect managerial turnover, creating selection into treatment implementation. If managers in treated municipalities exit or enter at higher rates, estimated treatment effects might partly capture changes in implementer composition rather than behavioral responses to the contract. However, Table ?? shows that the probability a manager remained in the municipality throughout the RCT period is statistically indistinguishable across treatment arms, mitigating concerns about differential turnover.

²⁴The endline survey was completed by around 40% of the universe of health facilities located in the districts that are part of the RCT. This survey take-up rate is broadly in line with meta-analytic benchmarks for online surveys (Wu et al., 2022). I was not able to collect baseline information on health facilities. In Table ??, I use health facilities' predetermined characteristics that should not have been affected by the intervention to test for balance in covariates. None of the differences in variables is statistically significant, confirming that health facilities in districts under the input-based contract arm constitute a valid counterfactual for those in districts under mixed incentive districts.

partial-bonus threshold is the minimum performance level a municipality must reach to receive 75% of the bonus tied to that indicator. The full-bonus threshold is the performance level at which the entire bonus is paid. Municipalities performing below the partial-bonus threshold receive nothing for that indicator. Performance above the full-bonus threshold is feasible, but generates no additional payment. Both thresholds are set in advance based on each municipality's baseline performance, are common across treatment arms, and are not modified by the assignment of the incentive contract.

Formally, each municipality d is eligible for a total potential bonus B_d , distributed equally across K performance indicators, so that each indicator k carries a potential bonus of $B_{dk} = B_d/K$. Municipality d 's payment for indicator k , denoted as P_{dk} , follows the step function below, where x_{dk} denotes performance, $\underline{x}_{dk}$ denotes the partial-bonus threshold, and $\bar{x}_{dk}$ denotes the full-bonus threshold:

$$P_{dk} = \begin{cases} 0, & \text{for } x_{dk} < \underline{x}_{dk} \\ 0.75 * B_{dk}, & \text{for } \underline{x}_{dk} \leq x_{dk} < \bar{x}_{dk} \\ B_{dk}, & \text{for } x_{dk} \geq \bar{x}_{dk} \end{cases} \quad (1)$$

Figure ?? plots the payment schedule for a single indicator following Equation 1. The municipal incentive payment $P_d = \sum_{k=1}^K P_{dk}$ is disbursed after performance verification and at the beginning of the following fiscal year, without affecting the ongoing implementation. Before the trial began, from January to July 2024, municipalities were evaluated solely on whether they met the predetermined target for the indicator measuring the share of children under 12 months who received a home visit. Consequently, the total bonus that municipalities could obtain depended entirely on their performance in that single indicator.

5.2 The Intervention

A new incentive scheme for the period August – December 2024 was developed in joint collaboration with the Ministry of Economy and the Ministry of Health. It was formalized through a series of legal resolutions.²⁵ The incentive scheme considers up to three indicators:²⁶

- (i) The percentage of children under 12 months old who receive home visit

²⁵See, among others: [Law 31953](#), [DS 318-2023-EF](#), [RD 0006-2024-EF/50.01](#), [DS 132-2024-EF](#), and [RD 0027-2024-EF/50.01](#).

²⁶The legal norm containing the details of these indicators can be found [here](#).

- (ii) The percentage of pregnant women who receive home visits
- (iii) The percentage of children who are non-anemic or have not experienced a decline in hemoglobin levels between their six- and twelve-month blood screenings

Based on these indicators, municipalities were offered one of two alternative incentive contracts:

- Under the *input-based incentive contract*, municipal performance is evaluated exclusively on the two input indicators (i and ii). Municipalities are eligible for a total potential bonus of B_d , which is divided equally across the two indicators, so that each indicator carries a potential bonus of $B_{dk} = B_d/2$.
- Under the *mixed incentive contract*, municipal performance is evaluated on the same two input indicators (i and ii) as well as the outcome indicator (iii). The total potential bonus B_d is held constant across contracts but is divided equally across the three indicators, such that each indicator carries a potential of $B_{dk} = B_d/3$. Critically, the mixed incentive did not expand the total financial stakes faced by municipalities — it only reoriented them, reducing the share of the bonus tied to each input indicator while introducing an additional outcome-based component.

5.3 Manager Motivation and Perceived Stakes

Understanding how managers perceive the incentive scheme is important for interpreting their behavioral responses to the two contracts. A natural question is why managers respond to it at all, given that the bonus accrues to the municipality rather than to them personally and arrives with a lag of roughly a year. Table 1 presents descriptive statistics from the managers' baseline survey on three dimensions relevant to this question: their beliefs about the cost of missing performance targets, their beliefs about how their own effort maps into children's health, and their underlying motivation.

On the first, managers anticipate substantial personal consequences from missing their targets, elicited as perceived probabilities at baseline: 56% believe they would be dismissed and 31% expect reassignment to another department, and they perceive these stakes as cascading down the team, assigning a 37% probability that CHWs would also face dismissal. On the second, managers see their own effort as decisive: 87% report that program success depends on what they do. They

are also intrinsically motivated: 55% cite population wellbeing rather than target attainment as their strongest motivation. Altogether, these patterns indicate that managers respond to the incentive not because of the bonus itself, but because of the personal accountability the scheme creates and their intrinsic motivation to improve child health.

Table 1: Managers' Beliefs, Motivation, and Perceived Stakes

	Total (%)	Urban (%)	Rural (%)
Number of districts	877	245	632
Manager's primary motivation is population wellbeing	58.24	66.94	54.85
Manager's effort is key to program success	86.31	83.81	87.29
Home Visit is the most effective strategy to prevent anemia	47.37	55.92	44.04
<i>Perceived consequences of missing performance targets</i>			
Manager dismissed	59.11	66.26	56.32
Manager reassigned to another department	30.06	26.85	31.31
Supervisors dismissed	54.25	65.70	49.79
CHWs dismissed	37.48	41.49	35.92
<i>Perceived key barrier to anemia recovery</i>			
Biology	45.88	40.82	47.85
Economy	30.89	31.84	30.52
Culture	12.13	15.10	10.97
Other barrier	11.10	12.24	10.65

Notes: *Manager's primary motivation is population wellbeing* is a dummy equal to 1 if the manager reports that contributing to population wellbeing is their primary motivation for implementing the program. *Manager's effort is key to program success* is a dummy equal to 1 if the manager reports that program success depends on what they do. *Home Visit is the most effective strategy to prevent anemia* is a dummy equal to 1 if the manager indicates that home visits is the most effective strategy to prevent anemia over the alternative strategies offered. *Manager dismissed*, *Manager reassigned*, *Supervisors dismissed*, and *CHWs dismissed* are the expected probabilities that the respective position would face that consequence should the municipality fail to meet all performance indicators. *Perceived key barrier to anemia recovery* reports the share of managers selecting each option; the categories are mutually exclusive. Data come from the managers' baseline survey.

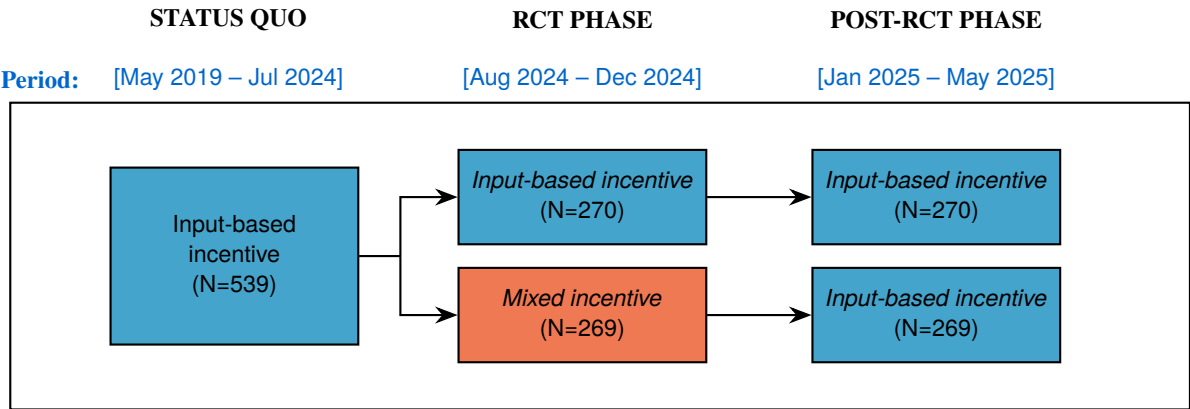
5.4 Randomization and Econometric Specification

I use a clustered randomized controlled trial to evaluate the impacts of the mixed incentive on municipality's service delivery and managerial practices. Randomization was done on the group of rural municipalities at the district level.²⁷ For evaluation purposes, I restrict the sample to rural

²⁷All the 245 urban municipalities were assigned to the mixed contract by government's decision.

municipalities with at least 15 children, resulting in a final group of 539 municipalities. Using a random number generator in STATA, 270 were randomly assigned to input-based contracts, while 269 received an input- and outcome-based incentive contract or “Mixed incentive”. Figure ?? shows the geographical distribution of the municipalities based on their involvement by the Home Visit Program and by treatment arm.

Figure 3: Experimental Design



Notes: This figure illustrates the experimental design. In the status quo (May 2019 to July 2024), all municipalities used to be incentivized only by the provision of service delivery (i.e., home visit coverage). During the RCT phase (August to December 2024), municipalities were randomly assigned to either continue with the same scheme (N=270) or to switch to a mixed incentive during the RCT phase (N=269). In December 2024, an unanticipated administrative delay at the central government prevented the renewal of the mixed incentive contract. Therefore, all municipalities reverted to the input-based scheme between January and May 2025 (post-RCT phase).

The treatment assignment was disseminated through multiple institutional channels to ensure transparency and municipal awareness. In late July 2024, the Ministry of Economy and Finance published the legal resolutions in the national newspaper and on its website. These resolutions included a list of all participating municipalities along with the indicators and performance thresholds assigned to each. The Ministry of Health complemented this announcement with in-person and virtual workshops for municipal authorities. To facilitate clearer communication, and in close coordination with the Ministry of Health, I implemented a nationwide online information campaign during September 2024. Unlike the consolidated official list (Figure ??), this campaign delivered municipality-specific information, providing each municipality only with the indicators and thresholds relevant to its own performance evaluation. A timeline of these dissemination activities is presented in Figure 2, and additional details on the information campaign are provided in Appendix ??

For outcomes at the children level, I estimate the following specification using ordinary least

squares (OLS):

$$Y_{id} = \alpha + \beta MixedIncentive_d + Y_{id}^0 + \epsilon_{id} \quad (2)$$

where Y_{id} is the outcome of interest at follow-up for child i in municipality d at baseline. $MixedIncentive_d$ is an indicator equal to 1 if municipality d was assigned to the treatment group (i.e. mixed incentive contract) and 0 otherwise. The comparison group represents the status quo scenario with municipalities under the input-based incentive contract. Because municipalities were randomly assigned to the mixed-incentive and input-based contracts, the input-based group constitutes a valid counterfactual for the mixed-incentive group by design, so β identifies the causal effect of the mixed-incentive contract. Since the randomization was conducted at the district level, I cluster the standard errors by district. Y_{id}^0 is the baseline level of the outcome variable and it is included when available to improve precision.

I estimate the same specification at two additional levels of aggregation, depending on the data source for each outcome. For outcomes from the manager survey and for the administrative measures of home visit coverage, staff and budget, the unit of observation is the municipality. Each municipality designates one manager for the Home Visit Program, so the manager-level and the municipality-level samples coincide. I estimate:

$$Y_d = \alpha + \beta MixedIncentive_d + \gamma Y_d^0 + \epsilon_d \quad (3)$$

where Y_d is the follow-up outcome for municipality d and Y_d^0 is its baseline value. I surveyed managers before the intervention started, so I observe baseline levels for most outcomes and include Y_d^0 when it is available.

For outcomes from the health facility survey, the unit of observation is the health facility h located in municipality d . I estimate:

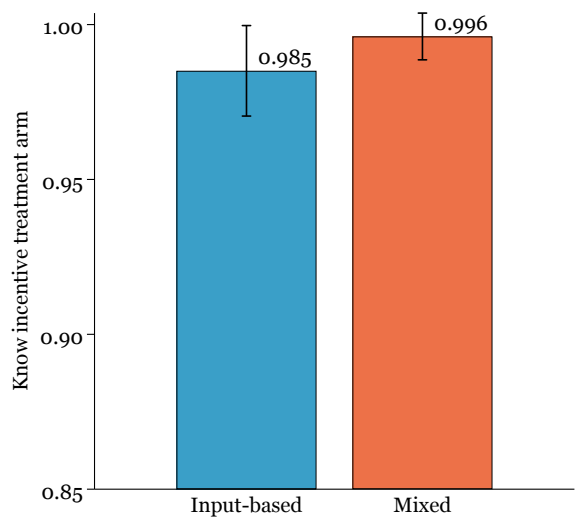
$$Y_{hd} = \alpha + \beta MixedIncentive_d + \epsilon_{hd} \quad (4)$$

I collected health facility data only at endline, so I cannot control for baseline outcomes in these regressions. Across all three levels, treatment varies only at the municipality level, so I cluster the standard errors by municipality throughout.

5.5 Incentive Awareness and Participation

Before examining behavioral responses, it is essential to confirm that municipalities correctly understood the performance criteria associated with their assigned contracts and chose to participate in the incentive scheme. Participation in the Home Visit Program was near-universal during the RCT period: 99.6% of municipalities conducted at least one home visit.²⁸ Regarding comprehension, approximately two months after the intervention began, 98.2% of municipalities correctly identified whether they were under an input-based or a mixed incentive contract (Figure 4).²⁹ This supports the internal validity of subsequent analyses.

Figure 4: Knowledge of Treatment Status



Notes: This graph shows the proportion of managers who correctly identified the indicators on which their municipality would be evaluated under the incentive scheme. The information is based on the Midline Survey to Managers.

5.6 Balance Checks

Table 2 presents the balance tests comparing districts assigned to the input-based incentive and those assigned to the mixed incentive at baseline. Panel A describes district-level socioeconomic

²⁸Participation is verified using the Home Visit Program’s app, through which CHWs register completed visits in real time. Two municipalities in the input-based incentive arm had no target children in their jurisdiction during the RCT period and were therefore unable to conduct home visits; all remaining municipalities had at least one home visit recorded.

²⁹Municipal managers also demonstrated relatively strong comprehension of the definition and objective of the anemia indicator (Table ??). Approximately 81% correctly recognized that the outcome indicator is a composite measure — namely, that performance improves when a child either remains non-anemic or exhibits an increase in hemoglobin levels between 6 and 12 months of age. In contrast, understanding of the more technical components of the indicator, such as the exact rules governing the numerator, denominator, and deductions, was substantially lower, indicating limited grasp of how performance would ultimately be evaluated.

525 conditions. On average, 20% of households are connected to a public water and sewage network,
and 83% of the population is covered by government-sponsored health insurance. Population (18
or more years old) has approximately 7.2 years of education. Municipalities, on average, spend
an average of USD 606.7 per capita on total expenditure, USD 16.2 per capita on health, and
USD 6.1 per capita on nutrition.

530 Panel B describes the health facilities operating in each district. Districts contain 5.6 health
facilities on average. Half of these facilities are Type I-1 and participate in the HV Program.
These health facilities sit approximately 15.99 kilometers from the district capital on average,
and have operated for 27.77 years.

535 Panel C summarizes key attributes of the Home Visit Program's workforce. Managers are,
on average, 33 years old, 44% are male, and 30% has higher education. They bring an average of
8.6 years of work experience, including about 1.9 years specifically in the Home Visit Program.
Supervisors constitute a relatively small team, with an average of 0.6 supervisors per municipality,
and 29% have a higher-education degree. Community Health Workers form the largest group,
with an average of 11.6 CHWs per municipality; about 16% of them have completed higher
540 education. Panel D describes baseline child health indicators. On average, children received 5.3
health facility visits and their caregivers collected iron supplements approximately 5.5 times.
12% percent of children were anemic at either 6 or 12 months of age.

Regarding balance across treatment arms, the only statistically significant difference arises in
the gender composition of managers: municipalities in the mixed incentive have a higher share of
545 male managers ($p < 0.01$).³⁰ Overall, fewer than 10% of the differences across all variables in the
table are statistically significant, reinforcing that the random assignment generated well-balanced
treatment arms at baseline.

³⁰*A priori*, it is unclear whether—and in which direction—this difference could bias the estimates, as the relationship between manager gender and managerial practices is not well established. However, robustness checks that control for manager gender yield virtually identical results, indicating that this imbalance does not affect the findings.

Table 2: Balance at Baseline

	Input-based incentive mean	Mixed incentive mean	Diff (SE)	p-value of Diff
<i>Number of Districts</i>	270	269	-	-
<i>Number of Health facilities</i>	1,507	1,494	-	-
<i>Number of Children</i>	7,303	6,501	-	-
<i>Panel A: District Characteristics</i>				
Population has government-sponsored health insurance	0.82	0.83	0.02 (0.01)	0.08*
Years of education (adults 18+)	7.25	7.21	-0.04 (0.07)	0.56
Total expenditure (USD per capita)	556.05	657.61	101.56 (66.15)	0.13
Expenditure in health (USD per capita)	17.17	15.28	-1.90 (5.80)	0.74
Connected to public water and sewage network	0.479	0.521	0.042 (0.044)	0.339
Poverty rate	0.528	0.487	-0.042 (0.044)	0.340
<i>Panel B: Health Facility Characteristics</i>				
Number of HF	5.581	5.554	-0.028 (0.441)	0.950
<i>Health Facility Type</i>				
Type I-1: No physician post	0.486	0.509	0.022 (0.027)	0.404
Type I-2: Physician post	0.341	0.324	-0.017 (0.027)	0.537
Type I-3: Polyclinic	0.148	0.139	-0.009 (0.019)	0.652
Type I-4: Inpatient facility	0.024	0.028	0.003 (0.008)	0.666
Distance from Type I-1 health facility to capital (km)	13.340	12.700	-0.640 (1.970)	0.745
Years since Type I-1 health facility opened	29.589	22.398	-7.191 (3.093)	0.021
<i>Panel C: Home Visit Program Characteristics</i>				
<i>Managers' Characteristics</i>				
Age	33.18	33.26	0.08 (0.70)	0.91
Dummy if male	0.38	0.50	0.12 (0.04)	0.01***
Has higher education	0.30	0.29	-0.01 (0.04)	0.87
Years of experience in the HV Program	1.81	1.94	0.12 (0.15)	0.40
<i>Supervisors' Characteristics</i>				
Number of supervisors	0.65	0.57	-0.08 (0.14)	0.58
Has higher education	0.28	0.30	0.02 (0.04)	0.69
<i>Community Health Workers' Characteristics</i>				
Number of CHW	11.08	12.05	0.97 (1.13)	0.39
Has higher education	0.16	0.15	-0.01 (0.02)	0.79
<i>Panel D: Children characteristics</i>				
Number of HF visits	11.85	12.61	-0.62 (0.40)	0.12
Number of iron pick-ups	2.38	2.56	-0.13 (0.10)	0.18
Number of vaccine doses	5.46	5.70	-0.14 (0.12)	0.23

Notes: Panel A reports district-level socioeconomic characteristics. Panel B reports characteristics of the health facilities located in each district, including facility type, distance to the district capital, and years of operation. Panel C summarizes characteristics of managers, supervisors, and Community Health Workers involved in the Home Visit Program. Panel D presents baseline child health and service-use indicators. All variables are measured at baseline. Differences come from a regression of each variable on the treatment indicator, with standard errors clustered at the district level reported in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

6 Results During the RCT Period

6.1 Effects on Home Visit Coverage

Before the RCT, anemia rates remained high even though home-visit coverage was near-universal and municipalities routinely met the visit thresholds. Policymakers traced the problem to the quality of the visits, not their quantity. Outcome-based incentives were therefore meant to shift effort along the quality margin rather than to expand coverage.³¹ Consistent with this, Table 3 shows that the mixed-incentive contract did not affect home visit coverage for either children or pregnant women. For children, the estimated effects on both overall coverage and target achievement are small and statistically indistinguishable from zero (0.5 and -0.4 p.p.) relative to the control mean of 88% and 75%, respectively. The same pattern holds for pregnant women. Coverage remains around 94% in both arms and the likelihood of meeting the partial-bonus target is similarly unaffected.

Table 3: Effects on home visit coverage

	Mixed incentive	Control mean	Effect (%)	<i>N</i>
Children: visit coverage	0.005 (0.009)	0.880	0.6	539
Children: met partial-bonus threshold	-0.004 (0.021)	0.748	-0.5	539
Pregnant women: visit coverage	-0.008 (0.015)	0.941	-0.9	539
Pregnant women: met partial-bonus threshold	-0.004 (0.028)	0.878	-0.5	539
Outcome indicator	0.011 (0.011)	0.811	1.4	539

Notes: Estimates come from district-level regressions. Standard errors clustered at the district level are in parentheses. *Visit coverage* is the share of eligible children or pregnant women who received timely home visits; *met partial-bonus threshold* equals 1 if the municipality reached at least the partial-bonus threshold for that indicator. The children's specifications control for the baseline value of the dependent variable; the pregnant women's specifications do not, as no baseline is available. No other controls are included. Statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ (unadjusted); ★ $p < 0.1$, ★★ $p < 0.05$, ★★★ $p < 0.01$ (MHT-adjusted). All outcomes belong to a single MHT family.

³¹Baseline coverage was already near its maximum. The program locates target families from administrative records, using the latest address reported by any public service provider who attended the child. CHWs try to trace unreachable households, but outdated addresses and recent migration leave a residual fraction unlocatable. This limits the scope for further coverage gains.

6.2 Effects on Managerial Practices

This subsection examines the impact of the mixed-incentive contract on the managerial practices described in Section 3. Table 4 reports the results in five panels, one per practice, drawing on the managers and health facilities' surveys.³² Panel A covers the two components of monitoring: tracking the performance indicators on which the municipality is evaluated, and verifying CHWs' visits. Panel B covers case management, from identification through coordination with health facilities. Panels C and D cover motivation of CHWs and training.

Managers intensified monitoring, but only along the indicator-tracking margin. The first row of Panel A shows that managers under the mixed contract were 13.7 p.p. more likely to track the anemia indicator daily or biweekly, an 18.9% increase over the input-based group. This is the indicator their bonus depended on. The second row shows no corresponding change in direct verification of CHWs' visits. Managers were no more likely to monitor CHWs on a weekly or biweekly basis (i.e., accompany CHWs to observe visit quality first-hand, call households, re-visit, or use any other method to confirm a visit took place).

Referral management covers the processes managers use to identify children at risk, initiate referrals, and follow them through to treatment. Panel B shows the effects of the mixed incentive contract on it. Managers under the mixed contract were 12.4 p.p. more likely to strengthen care for children identified in the field as at risk of anemia, a 24.4% increase. The response extended to the health facility. Managers were 19.5 p.p. more likely to coordinate referrals directly with facility staff to identify open slots and arrange appointment days for referred children (83% increase). In the standard process, a mother travels to the facility, queues to register an appointment, and returns on a later date to attend it. Coordination lowers the cost families face in reaching a medical consultation. In addition to that, CHWs went further outside of their duties and accompanied mothers to the health facility to ensure they do not miss their appointment, a 55.2% increase over a control mean of 0.26.³³

Panel C turns to how managers motivated CHWs. Managers under the mixed contract were 17.1 p.p. more likely to reward CHWs, a 46.1% increase. The rewards were non-monetary and not tied to performance (i.e., public recognition, training, career advancement opportunities, and in-kind goods). Panel D reports the effects on training. Municipalities under the mixed contract

³²Results are robust to alternative samples and levels of aggregation (Table ??). Estimates from the managers' survey hold when the sample is restricted to districts with at least one surveyed health facility (Panel A), and estimates from the health facilities' survey hold when the data are aggregated to the district level (Panel B).

³³CHWs are volunteers whose transport stipend covers up to three home visits per month, and it does not cover this additional trip.

coordinated with health facilities to run more sessions. CHWs received 1.2 additional sessions from facility staff; however, this estimate is significant only at the 10% level.

Table 4: Effects on Managerial Practices During the RCT Period

	Mixed incentive	Control mean	Effect (%)	N
Panel A. Monitoring				
Monitored anemia KPI indicator	0.137*** (0.038)	0.723	18.949	442
Monitored CHWs daily or biweekly	0.046★ (0.036)	0.589	7.810	442
Panel B. Referral management				
Improved care for children at risk of anemia	0.124*** (0.047)	0.509	24.361	442
Referral is coordinated with health facility	0.195*** (0.052)	0.235	82.979	397
CHWs take mothers to health facility	0.144** (0.058)	0.261	55.172	397
Panel C. Motivation of CHWs				
CHW was incentivized	0.171*** (0.047)	0.371	46.092	442
Panel D. Training CHWs				
Number of CHW training sessions	1.194** (0.697)	3.797	31.446	397

Notes: Estimates come from district-level and health facility-level regressions. Standard errors clustered at the district level in parentheses. Outcomes in Panels A and C and in the first row of Panel B come from the managers' survey and are measured at the district level ($N = 442$); the remaining rows come from the health facilities' survey and are measured at the facility level ($N = 397$). *Effect (%)* reports the point estimate as a percentage of the control mean. Only variables in Panel A control for the baseline value of the dependent variable; the remaining specifications do not, as no baseline is available. No other controls are included. Statistical significance: $*p < 0.1$, $**p < 0.05$, $***p < 0.01$ (unadjusted); $\star p < 0.1$, $\star\star p < 0.05$, $\star\star\star p < 0.01$ (MHT-adjusted). All outcomes belong to a single MHT family.

Importantly, these managerial responses operated entirely through behavioral changes rather than resource expansion. Table ?? reports effects on the Home Visit program's staffing levels and composition, for both CHWs and supervisors. Municipalities in the mixed-incentive arm employed no more community health workers or supervisors than municipalities under the input-based contract. The share of either group holding a higher-education degree did not change either. Managers thus did not expand or upgrade their workforce. Municipal expenditures corroborate this pattern (Table ??). Total municipal spending and total health spending show no significant effect (Panel A). The same holds when total health spending is broken into its four components (Panel B), covering payments to CHWs and other Home Visit Program spending,

infant and maternal health spending outside the Program, and spending on other health conditions.

6.3 Effects on Service Utilization and Health Outcomes

The analysis sample consists of children who were between 1 and 9 months old during the five-month RCT phase. This age window ensures two things: first, that all children in the sample were exposed to the Home Visit Program for at least one month before their first routine anemia screening; and second, that all children reached the age for that first screening during the RCT phase (i.e., 6 to 8 months). This cohort is also particularly well suited for evaluating impacts on health service demand and health outcomes, as most routine preventive services — including vaccinations, growth-monitoring visits, and iron supplementation — are delivered between 1 and 9 months of age.³⁴

Table 5 reports the effects of the mixed-incentive contract on health service utilization (Panel A) and children health outcomes (Panel B) during the RCT period. Panel A shows statistically significant increases in the total number of health facility visits, iron supplement pick-ups, and vaccine doses received by children. Relative to the input-based contract, the mixed-incentive scheme increased health facility visits by 0.5 visits (8.2% relative to the control mean), iron supplement pick-ups by 0.24 pick-ups (112%), and vaccine doses by 0.27 doses (14.2%). These findings indicate that parents used preventive health services more intensively, consistent with CHWs becoming more effective at encouraging service uptake and facilitating referrals to health facilities.

Panel A also reports the effects on the probability that a child received a blood test for anemia screening and on health outcomes. There are no effects of the mixed incentive on blood testing. Because hemoglobin levels and anemia diagnoses are only observed for tested children, this null result is important for the interpretation of the remaining variables. The absence of an effect on blood testing suggests that the mixed-incentive contract did not differentially affect selection into anemia screening. Consequently, estimated impacts on hemoglobin and anemia outcomes are unlikely to be driven by changes in testing behavior and instead reflect changes in underlying health status.

With respect to health outcomes, the estimated impacts are small in magnitude and statistically indistinguishable from zero across all indicators (Panel B). There is no significant impact on

³⁴According to the national immunization schedule, 12 of the 17 vaccine doses administered during the first year of life are delivered between 1 and 9 months of age (Figure ??).

hemoglobin concentration or on the probability of being classified as anemic. The estimated effect on hemoglobin is close to zero and precisely estimated: the point estimate is 0.033 g/dL (s.e. 0.070), which allows to rule out effects larger than approximately 0.10 g/dL. This bound is less than one-fifth of the 0.56 g/dL that clinical trials achieve with enforced adherence (Section 2). The effect on anemia prevalence is likewise tightly centered at zero, with a point estimate of 0.001 (s.e. 0.022) against a control mean of 0.202, ruling out changes larger than 4.4 percentage points. The effect on weight-for-age z-scores is similarly null, ruling out improvements larger than 0.11 standard deviations. The same pattern holds among children old enough to have been potentially exposed to the Home Visit program for at least three months (Table ??).

Since maternal nutrition and sanitation shape what supplementation can achieve, I examine heterogeneity by baseline district-level poverty and by access to drinking water and sanitation. I find no evidence of differential treatment effects along either dimension (Tables ?? and ??), suggesting that the absence of average effects does not appear to be driven by variation in these baseline conditions.

Finally, I follow the same children into the five months after the RCT ended, when every municipality operated under an input-based contract (Table ??). The utilization gains persist. Children in mixed-incentive districts received 0.59 more health facility visits (5.0%) and 0.38 more iron pick-ups (16.0%), though the vaccine effect disappears. Health outcomes remain null. The hemoglobin estimate is 0.115 g/dL (s.e. 0.126), the anemia estimate is -0.033 (s.e. 0.038), and the weight-for-age estimate is -0.006 (s.e. 0.039). Fewer children were screened during this period, so the hemoglobin bound widens to 0.36 g/dL, still well below the 0.56 g/dL that supervised trials achieve. Five additional months of follow-up therefore leave the health null intact.

Table 5: Effects on Health Service Utilization and Health Outcomes during the RCT Period
(Children exposed for at least 1 month during the RCT period)

	Mixed incentive	Control mean	Effect (%)	<i>N</i>
Panel A. Health service utilization				
Number of HF visits	0.367** (0.172)	9.514	3.9	13,804
Number of iron pick-ups	0.355** (0.174)	0.362	98.1	13,804
Number of vaccine doses	0.283* (0.165)	3.215	8.8	13,804
Panel B. Health outcomes				
Weight-for-Age Z-score (WAZ)	0.053 (0.066)	0.439	12.1	12,505
Took blood test	0.020* (0.012)	0.857	2.3	13,804
Anemic	0.003 (0.020)	0.195	1.5	11,573
Hemoglobin (g/dL)	-0.034 (0.069)	12.600	-0.3	11,573

Notes: This table reports the effects of the mixed-incentive contract on health service utilization and child health outcomes for the children who were exposed at least 1 month to the intervention during the RCT period and before the blood testing. Exposure duration is determined by birth cohort. Each row is a separate regression estimated at the child level. Panel A reports the total number of health facility (HF) visits, iron supplement pick-ups, and vaccine doses received during the study period. According to the national immunization schedule, children aged 2–12 months are scheduled to receive up to 17 routine vaccine doses. In Panel B, *Took blood test* indicates whether the child received a blood test for anemia screening; *Hemoglobin (log)* is the log of hemoglobin concentration measured during routine anemia screening; *Anemic* equals one if the child is classified as anemic based on altitude-adjusted hemoglobin thresholds. Effect (%) reports the point estimate as a percentage of the control mean. All estimates control for the outcome at baseline, except *Took blood test*. Cohort fixed effects are included throughout, and the *Anemic* regression additionally includes linear and quadratic controls for altitude. Standard errors clustered at the district level are in parentheses. Statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ (unadjusted); ★ $p < 0.1$, ★★ $p < 0.05$, ★★★ $p < 0.01$ (MHT-adjusted). When correcting the p -values for multiple hypothesis testing, I group the outcomes in two MHT families, one per panel.

Table 6: Effects on Health Service Utilization and Health Outcomes during the Post RCT Period
(Children exposed for at least 1 month during the RCT period)

	Mixed incentive	Control mean	Effect (%)	N
Panel A. Health service utilization				
Number of HF visits	0.526* (0.290)	9.514	5.5	13,804
Number of iron pick-ups	0.419* (0.238)	0.362	115.7	13,804
Number of vaccines doses	0.070 (0.045)	3.215	2.2	13,804
Panel B. Health outcomes				
Weight-for-Age Z-score (WAZ)	0.017 (0.045)	0.425	4.0	12,952
Took blood test	0.020* (0.011)	0.624	3.2	13,804
Anemic	-0.009 (0.021)	0.265	-3.4	8,493
Hemoglobin (g/dL)	0.047 (0.058)	12.266	0.4	8,493

Notes: This table reports the effects of the mixed-incentive contract on health service utilization and child health outcomes for the children who were exposed at least 1 month to the intervention before the blood testing. Exposure duration is determined by birth cohort. Each row is a separate regression estimated at the child level. Panel A reports the total number of health facility (HF) visits, iron supplement pick-ups, and vaccine doses received during the study period. According to the national immunization schedule, children aged 2–12 months are scheduled to receive up to 17 routine vaccine doses. In Panel B, *Took blood test* indicates whether the child received a blood test for anemia screening; *Hemoglobin (log)* is the log of hemoglobin concentration measured during routine anemia screening; *Anemic* equals one if the child is classified as anemic based on altitude-adjusted hemoglobin thresholds. *Effect (%)* reports the point estimate as a percentage of the control mean. Estimates control for the outcome at baseline, except *Took blood test*. Cohort fixed effects are included throughout, and the *Anemic* regression additionally includes linear and quadratic controls for altitude. Standard errors clustered at the district level are in parentheses. Statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ (unadjusted); ★ $p < 0.1$, ★★ $p < 0.05$, ★★★ $p < 0.01$ (MHT-adjusted). When correcting the p -values for multiple hypothesis testing, I group the outcomes in two MHT families, one per panel.

7 Results During the Post RCT Period

The RCT incentive scheme was originally designed to continue for a second semester beyond December 2024. However, an unexpected administrative complication in the publication of the legal norms regulating the new incentive period prevented its renewal, and all municipalities reverted to the input-based incentive scheme from January to May 2025. This return to the status quo was unanticipated at all levels — neither the central government nor the municipalities had foreseen it. I use the five months following the end of the RCT to examine whether the changes in managerial practices and health service utilization induced by the mixed incentive persisted after its removal, and whether they translated into detectable improvements in child health outcomes.

7.1 Persistence of Managerial Practices

Table 7 repeats the specifications of Table 4 for the five months after the mixed contract ended. Managers did not revert once the outcome indicator disappeared from their contract. Panel A shows they remained 13.4 p.p. more likely to track the anemia indicator, against 13.7 p.p. during the RCT phase, and verification of CHWs' visits stays null, as before. Panel B shows the same pattern in referral management. Managers were still 11.0 p.p. more likely to strengthen care for children at risk and 10.3 p.p. more likely to coordinate with facility staff, and CHWs continued to accompany mothers (11.3 p.p.). Panel C shows managers still rewarding CHWs more (13.4 p.p.). Training shows no effects, as during the RCT phase. Staffing and municipal expenditure show no effects in this period either (Tables ?? and ??).

Table 7: Effects on Managerial Practices During the Post RCT Period

	Mixed incentive	Control mean	Effect (%)	<i>N</i>
<i>Panel A. Monitoring</i>				
Monitored anemia KPI indicator	0.134*** (0.042)	0.722	18.6	352
Monitored CHWs daily or biweekly	0.003 (0.044)	0.550	0.5	352
<i>Panel B. Referral management</i>				
Improved care for children at risk of anemia	0.110** (0.050)	0.606	18.2	352
Referral with HF is coordinated	0.103* (0.056)	0.205	50.2	237
CHWs take mothers to HF	0.113** (0.048)	0.154	73.4	237
<i>Panel C. Motivation of CHWs</i>				
CHW was incentivized	0.134** (0.053)	0.372	36.0	352
<i>Panel D. Training</i>				
Number of CHW training sessions	0.010 (0.714)	4.923	0.2	237

Notes: Estimates come from district-level and health facility-level regressions. Standard errors clustered at the district level in parentheses. Outcomes in Panels A and C and in the first row of Panel B come from the managers' survey and are measured at the district level ($N = 352$); the remaining rows come from the health facilities' survey and are measured at the facility level ($N = 237$). *Effect (%)* reports the point estimate as a percentage of the control mean. Only variables in Panel A control for the baseline value of the dependent variable; the remaining specifications do not, as no baseline is available. No other controls are included. Statistical significance: $*p < 0.1$, $**p < 0.05$, $***p < 0.01$ (unadjusted); $\star p < 0.1$, $\star\star p < 0.05$, $\star\star\star p < 0.01$ (MHT-adjusted). All outcomes belong to a single MHT family.

7.2 Longer-Run Effects on Service Utilization and Health Outcomes

Table 6 repeats the specifications of Table 5 for the post-RCT period. The health service utilization gains did not fade. Panel A shows that children in municipalities previously assigned to the mixed incentive attended 0.8 more health facility visits, a 13.8% increase over the control mean, against 3.9% during the RCT phase. Vaccine doses and blood testing show no differences across arms in this period. The absence of an effect on blood testing again suggests that the composition of tested children does not differ across arms, so the health estimates that follow are not driven by selection into screening. Health outcomes remain null. Panel B shows effects on hemoglobin, anemia prevalence, and weight-for-age z-scores that are small and statistically indistinguishable from zero, as during the RCT phase.

Table 8: Effects on Health Service Utilization and Health Outcomes during the Post RCT Period
(Children exposed for at least 1 month during the Post RCT period)

	Mixed incentive	Control mean	Effect (%)	N
Number of HF visits	1.139** (0.538)	0.805	141.5	11,633
Number of iron pick-ups	0.265 (0.216)	5.591	4.7	11,633
Number of vaccine doses	0.129 (0.152)	7.644	1.7	11,633

Notes: This table reports the effects of the mixed-incentive contract on health service utilization for the children who were exposed at least 1 month to the intervention during the Post RCT Period and before the blood testing. Exposure duration is determined by birth cohort. Each row is a separate regression estimated at the child level. Panel A reports the total number of health facility (HF) visits, iron supplement pick-ups, and vaccine doses received during the study period. According to the national immunization schedule, children aged 2–12 months are scheduled to receive up to 17 routine vaccine doses. In Panel B, *Took blood test* indicates whether the child received a blood test for anemia screening; *Hemoglobin (log)* is the log of hemoglobin concentration measured during routine anemia screening; *Anemic* equals one if the child is classified as anemic based on altitude-adjusted hemoglobin thresholds. *Effect (%)* reports the point estimate as a percentage of the control mean. Estimates control for the outcome at baseline, except *Took blood test*. Cohort fixed effects are included throughout, and the *Anemic* regression additionally includes linear and quadratic controls for altitude. Standard errors clustered at the district level are in parentheses. Statistical significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ (unadjusted); ★ $p < 0.1$, ★★ $p < 0.05$, ★★★ $p < 0.01$ (MHT-adjusted). When correcting the p -values for multiple hypothesis testing, I group the outcomes in two MHT families, one per panel.

8 Conclusion

This paper examines whether institutional incentive design can address the asymmetric information problem that plagues public service delivery in decentralized systems. Using a clustered randomized controlled trial across 539 rural municipalities in Peru, I compare an input-based contract with a mixed incentive contract that adds an outcome indicator tied to child anemia reduction. Municipalities assigned to the mixed incentive increased monitoring intensity, provided more discretionary rewards to Community Health Workers, took more active steps to manage high-risk cases, and strengthened coordination with health facilities — all without any increase in fiscal expenditure or staffing. This suggests that realigning managerial attention, rather than expanding resources, was the operative mechanism.

These managerial responses translated into measurable improvements in health service utilization. Children in mixed-incentive municipalities attended more health facility visits, collected iron supplements at higher rates, and received more vaccine doses during the intervention period. The breadth of these effects across multiple service dimensions points to a systematic improvement in community health outreach, consistent with CHWs becoming more effective at encouraging families to engage with the health system.

An unanticipated administrative delay in the publication of the legal norms governing the new incentive period caused all municipalities to revert to the status quo input-based scheme from January to May 2025, providing an opportunity to examine the durability of the mixed incentive's effects. The managerial changes induced during the RCT period proved largely persistent: municipalities previously exposed to the mixed incentive continued to monitor anemia indicators more closely, support high-risk children more proactively, and reward CHWs at higher rates even after the contract ended. Persistence was uneven across practices, however — relationship-based behaviors such as accompanying mothers to health facilities and coordinating referrals with health facility staff remained strong, while more resource-intensive efforts such as additional CHW training dissipated once the incentive was removed. These durable managerial changes continued to translate into higher health service utilization, with children in previously treated municipalities attending more facility visits and collecting iron supplements at significantly higher rates in the post-RCT period.

Despite higher health service utilization both during and after the intervention, the study produced no detectable improvements in child health outcomes over the full observation window. Anemia prevalence, hemoglobin levels, and weight-for-age z-scores remained statistically

indistinguishable from zero across all specifications, with confidence intervals tight enough to rule out effects of meaningful magnitude. A more plausible interpretation is that the biological pathways linking iron supplementation to hemoglobin recovery operate over longer horizons than the study window allows. While these findings suggest that contract design is a viable lever for improving implementation quality — and that its effects on managerial behavior and service utilization can outlast the contract itself — they also caution against expecting short-run health returns from interventions targeting conditions shaped by cumulative and slow-moving biological processes.

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